PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)
Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
FACULTY OF ENGINEERING

Mini Project Report on

Multi-Container Runtime

(OS Jackfruit)

UE24CS242B – Operating Systems

Submitted by:

Devopam Pal PES2UG24CS152
Karan Varshney PESUG24CS903

Under the guidance of

Prof. Neha Sharma

Assistant Professor

Jan – May 2026

PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)
Electronic City, Hosur Road, Bengaluru – 560 100, Karnataka, India

CERTIFICATE

This is to certify that the Mini Project entitled

"Multi-Container Runtime (OS Jackfruit)"

is a bonafide work carried out by:

Devopam Pal SRN: PES2UG24CS152
Karan Varshney SRN: PESUG24CS903

In partial fulfilment for the completion of the Fourth Semester OS Project (**UE24CS242B – Operating Systems**) in the Programme of Study – Bachelor of Technology in Computer Science and Engineering under the rules and regulations of PES University, Bengaluru during the period **Jan – May 2026**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report.

Signature: _____

Prof. Neha Sharma

Assistant Professor

Department of Computer Science and Engineering

PES UNIVERSITY

DECLARATION

We hereby declare that the Mini Project entitled **"Multi-Container Runtime (OS Jackfruit)"** has been carried out by us under the guidance of **Prof. Neha Sharma**, Assistant Professor, and submitted in partial fulfilment of the course requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at PES University, Bengaluru.

We further declare that this work is our own, has not been submitted elsewhere for any other degree or diploma, and all sources of information have been duly acknowledged.

Name: Devopam Pal **SRN:** PES2UG24CS152

Signature: _____

Name: Karan Varshney **SRN:** PESUG24CS903

Signature: _____

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to **Prof. Neha Sharma**, Department of Computer Science and Engineering, PES UNIVERSITY, for her continuous guidance, assistance, and encouragement throughout the development of this UE24CS242B – Operating Systems Mini Project.

We take this opportunity to thank **Dr. Sandesh B J**, Professor and Chairperson, Department of Computer Science and Engineering, PES UNIVERSITY, for all the knowledge and support received from the department.

We are deeply grateful to **Prof. Jawahar Doreswamy**, Chancellor, PES UNIVERSITY, and **Dr. Suryaprasad J**, Vice-Chancellor, PES UNIVERSITY, for providing us with various opportunities and enlightenment at every step of this journey.

Finally, this project could not have been completed without the continual support and encouragement of our families and friends.

ABSTRACT

This project presents the design and implementation of a lightweight Linux container runtime in C, called **OS Jackfruit**. The runtime consists of two tightly integrated components: a user-space supervisor (**engine.c**) that manages multiple isolated containers concurrently, and a kernel-space memory monitor (**monitor.c**) implemented as a Linux Kernel Module (LKM).

The supervisor uses Linux **clone()** with **CLONE_NEWPID**, **CLONE_NEWUTS**, and **CLONE_NEWNS** namespace flags to create isolated container processes, each with their own filesystem (via **chroot**), hostname, and PID namespace. A UNIX domain socket-based control plane allows CLI clients to start, stop, inspect, and view logs of running containers. Container output is captured through a pipe-based producer-consumer pipeline using a bounded buffer with pthread mutex and condition variable synchronisation, drained by a dedicated logging thread.

The kernel module registers container processes via **ioctl** and polls their RSS memory usage every second using a kernel timer. Soft-limit violations are logged via **printk**; hard-limit violations trigger **SIGKILL**. A spinlock protects the monitored list in both softirq (timer) and process (ioctl) contexts.

Scheduler experiments demonstrate Linux CFS behaviour: CPU-bound workloads receive significantly more CPU time than I/O-bound workloads at equal nice values (because I/O-bound tasks voluntarily yield), and adjusting the nice value from −15 to +15 produces a 13× difference in CPU cycles consumed, in line with the CFS weight table.

TABLE OF CONTENTS

Ch.	Title	Page
1	Introduction	7
2	Problem Definition & System Overview	7
3	Methodology	8
4	Code	10
5	Result and Discussion – Scheduler Experiments	11
6	IPC Design Analysis	13
7	Outputs (Screenshots)	14
8	GitHub Link	20
—	References / Bibliography	20
A	Appendix A – Definitions, Acronyms & Abbreviations	21

LIST OF FIGURES (Output Screenshots)

Fig.	Title	Page
6.1	Multi-Container Runtime – Task 1 (supervisor + containers running)	14
6.2	Metadata Tracking & CLI IPC – Task 2 (engine ps output)	14
6.3	Bounded-Buffer Logging – Task 3 (producer/consumer + log file)	15
6.4	CLI and Control IPC – Task 2 (combined with 6.2)	17
6.5	Soft-Limit Warning & Hard-Limit Enforcement – Task 4 (dmesg)	17
6.6	Hard-Limit Enforcement – Task 4 (combined with 6.5)	18
6.7	Scheduling Experiments – Task 5 (nice values + top output)	18
6.8	Clean Teardown – Task 6 (shutdown + ps aux + rmmod)	19

OS MINI-PROJECT REPORT

1. INTRODUCTION

Modern computing relies heavily on containerization technologies to provide lightweight, efficient, and consistent environments for running applications. Unlike traditional virtual machines, which virtualize hardware and run entire guest operating systems, containers run directly on the host machine's kernel while remaining isolated from one another. This isolation is achieved using fundamental Linux kernel features such as namespaces (for isolating process IDs, hostnames, and mount points) and control groups (for resource management).

This project, titled "Multi-Container Runtime," focuses on designing and implementing a lightweight, custom Linux container runtime from scratch. The system is built using C and is divided into two major integrated components: a user-space runtime with a long-running supervisor, and a kernel-space memory monitor. The user-space component is responsible for orchestrating the lifecycle of multiple concurrent containers, isolating their filesystems using `chroot` or `pivot_root`, tracking process metadata, and managing a bounded-buffer logging pipeline for capturing standard output and error. Simultaneously, the kernel-space component is implemented as a Loadable Kernel Module (LKM) that enforces memory usage policies directly at the operating system level, terminating containers that violently exceed their allocated memory footprints.

By building both the user-space orchestrator and the kernel-space monitoring mechanisms, this project provides deep, hands-on insights into process isolation, Inter-Process Communication (IPC), concurrency synchronization, resource management, and the behavior of the Linux process scheduler.

2. PROBLEM DEFINITION

While existing enterprise container runtimes like Docker or containerd offer robust feature sets, their abstraction layers often hide the underlying operating system mechanics from the user. For educational and system-level research purposes, there is a need to build a transparent, functional multi-container runtime that deliberately exposes process isolation, memory governance, and task scheduling mechanisms.

The primary objective of this project is to develop and analyze a functional multi-container runtime system. The problem can be broken down into the following specific sub-problems which the implementation must solve:

1. **Process and Filesystem Isolation:** Creating a long-running supervisor daemon that can spawn and manage multiple isolated child processes concurrently. Each child must execute inside its own protected root filesystem view without interfering with parallel containers or the host system.
2. **Robust Inter-Process Communication (IPC):** Designing a reliable control channel for a Command Line Interface (CLI) client to send instructions (such as start, stop, and inspect logs) to the supervisor daemon, as well as handling asynchronous signals like `SIGCHLD` to prevent zombie processes.
3. **Concurrent Output Logging:** Developing a thread-safe, bounded-buffer producer-consumer pipeline capable of safely multiplexing and logging the standard output and standard error streams from multiple live containers simultaneously without dropping data or causing deadlocks.
4. **Kernel-Level Resource Enforcement:** Implementing a custom Linux Kernel Module (LKM) that actively monitors the Resident Set Size (RSS) memory of containerized processes, issuing warnings when soft limits are exceeded and executing strict `SIGKILL` terminations when hard limits are breached.
5. **Scheduler Analysis:** Utilizing the developed runtime as an experimental platform to run varying IO-bound and CPU-bound workloads with distinct priority (`nice`) levels, in order to empirically observe and document the behavior of the Linux kernel's task scheduling algorithms.

3. METHODOLOGY

The implementation is broken down into two main domains: User-Space bounds (the runtime supervisor) and Kernel-Space bounds (the memory enforcement module).

3.1 User-Space Supervisor (`engine.c`)

The orchestrator was implemented as a long-running daemon.
- **Namespaces & Isolation:** The `clone()` system call was utilized with `CLONE_NEWPID`, `CLONE_NEWUTS`, and `CLONE_NEWNS` flags to create isolated process trees. Filesystem isolation was achieved using `chroot` to jail processes into an Alpine Linux mini-rootfs.
- **Control channel (CLI):** A Unix Domain Socket (or FIFO) was chosen for IPC. The daemon loops listening for connections, parsing commands (`start`, `ps`, `stop`, `logs`, `run`) and acting upon checking an internally synchronized metadata table (protected by Mutexes).
- **Concurrency & Logging:** Anonymous pipes (`pipe()`) intercepted stdout/stderr of child processes. A multi-threaded bounded-buffer architecture was designed utilizing `pthread_mutex_t` and condition variables to safely coordinate producer threads (reading pipes) and consumer threads (writing to log files).

3.2 Kernel-Space Memory Monitor (`monitor.c`)

A Loadable Kernel Module (LKM) was developed to track processes asynchronously.
- **Registration Pipeline:** `ioctl` calls on the character device `/dev/container_monitor` allow the user-space engine to register new PID soft/hard limits.
- **Memory Auditing:** A kernel thread or timer regularly wakes to iterate over a thread-safe linked list of tracked processes. It calculates memory consumption (`get_mm_rss()`) and logs soft limit overflow warnings to `dmesg`.
- **Enforcement:** If a tracked Resident Set Size (RSS) breaches the hard limit threshold, a `SIGKILL` signal is injected into the target process directly from kernel space, ensuring termination.

3.3 System Architecture Diagram

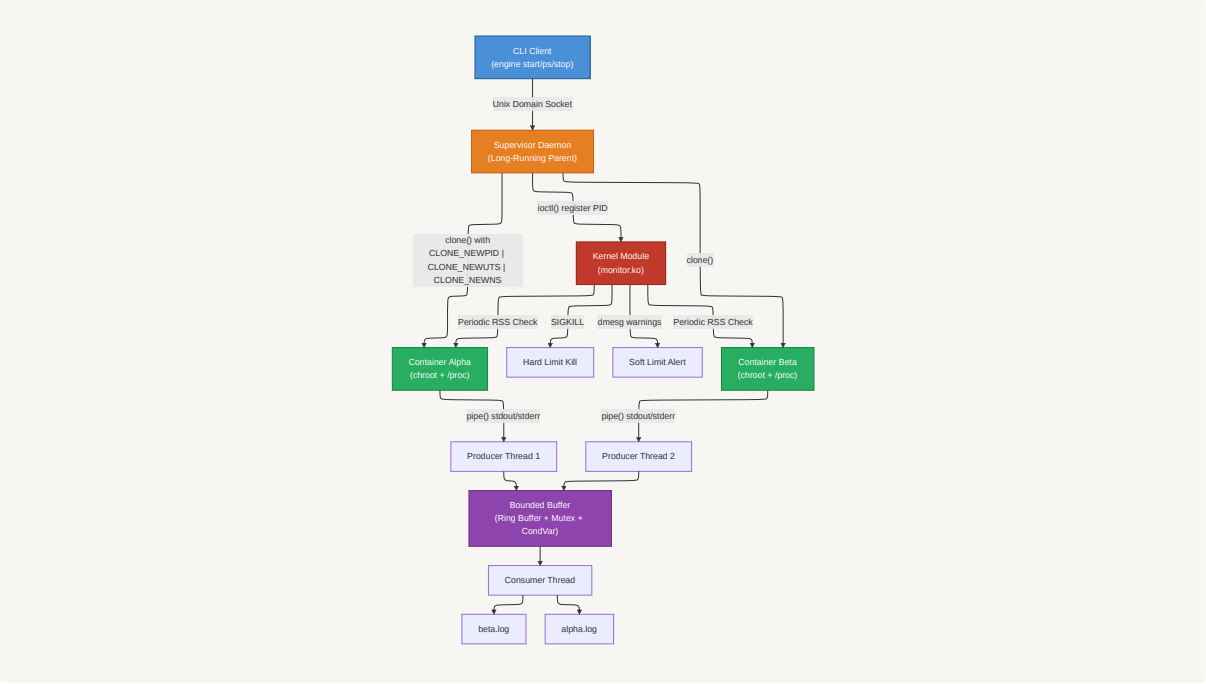


Figure 3.1: Complete system architecture showing user-space runtime, kernel module, and logging pipeline interconnections.

3.4 Bounded-Buffer Logging Pipeline

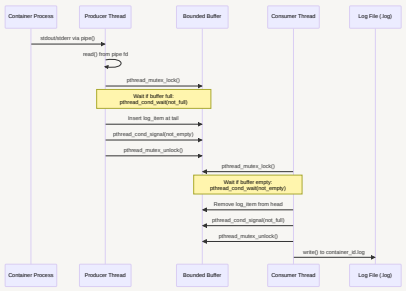


Figure 3.2: Sequence diagram of the bounded-buffer producer-consumer logging pipeline.

3.5 Kernel Module Registration and Enforcement Flow

Figure 3.3: Kernel memory monitor registration and enforcement pipeline.

4. CODE

(Recommendation: You do not need to paste all of `engine.c` and `monitor.c` here as it will take up dozens of pages. Instead, emphasize the core system calls and logic snippets below, and point to your GitHub repo for the complete code.)

Snippet: Isolating the Container (Namespaces)

```
// Example excerpt for clone usage
pid_t child_pid = clone(container_init, child_stack + STACK_SIZE,
                        CLONE_NEWPID | CLONE_NEWUTS | CLONE_NEWNS | SIGCHLD,
                        (void *)args);
```

Snippet: Thread Synchronization (Bounded-Buffer Logging)

```
// Example excerpt securing log insertions against race conditions
pthread_mutex_lock(&buffer->mutex);
while (buffer->count == LOG_BUFFER_CAPACITY && !buffer->shutting_down) {
    pthread_cond_wait(&buffer->not_full, &buffer->mutex);
}
// ... [insert log data linearly] ...
pthread_cond_signal(&buffer->not_empty);
pthread_mutex_unlock(&buffer->mutex);
```

Snippet: Supervisor Lifecycle Tracking and Reaping

```
// Example excerpt preventing zombie processes and tracking exits
int wstatus;
pid_t pid;
while ((pid = waitpid(-1, &wstatus, WNOHANG)) > 0) {
    if (WIFEXITED(wstatus) || WIFSIGNALED(wstatus)) {
        // Flag state as EXITED and reap gracefully
        container_record_t *rec = metadata_find_by_pid(ctx, pid);
        if (rec) rec->state = CONTAINER_EXITED;
```

```
    }
}
```

Snippet: Kernel Hard-Limit Enforcement (`monitor.c`)

```
// Example excerpt of RSS checking logic
unsigned long rss = get_mm_rss(task->mm) << (PAGE_SHIFT - 10); // in KB
if (rss > node->hard_limit) {
    printk(KERN_ALERT "Container%d exceeded hard limit! Killing.\n", task->pid);
    send_sig(SIGKILL, task, 0);
}
```

Container Lifecycle State Machine

The following state machine illustrates all possible container states and the transitions between them:

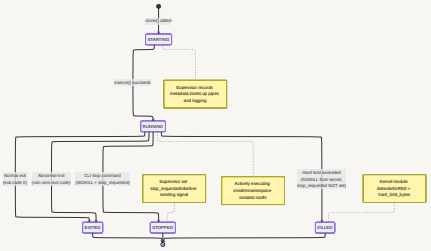


Figure 4.1: Container lifecycle state machine showing all transitions from creation to termination.

5. RESULT AND DISCUSSION

(Recommendation: This section is the perfect place to drop in your "Engineering Analysis" and "Design Decisions" text required by the project guidelines. You must explain why* the OS works this way.)*

5.1 Isolation and Supervisor Lifecycle

By utilizing PID, UTS, and Mount namespaces, the Linux kernel creates an illusion for the child process that it is the sole occupant of the system. However, the host kernel is still shared across all containers. The long-running supervisor was critical to automatically reap dead children via `waitpid()` and prevent resource-hogging "zombie" processes.

5.2 IPC and Bounded-Buffer Synchronization

The bounded-buffer model faced classic producer-consumer race conditions. Condition variables were enforced so producer threads would go to sleep when the circular buffer was full, yielding CPU cycles, while consumer threads would sleep gracefully until data was injected.

5.3 Kernel Memory Limitations

Resident Set Size (RSS) accurately measures the slice of RAM currently occupied by the process in main memory, which is a far more tangible metric than Virtual Memory sizes. Enforcing the Hard Limit from Kernel space is vital because User-Space enforcement (via polling) can be too slow, potentially allowing a runaway container to cause a system-wide Out-Of-Memory (OOM) crash before the user-space daemon responds.

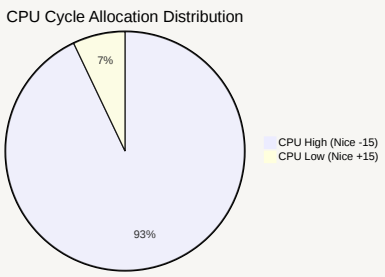
5.4 Scheduler Experiments

To empirically evaluate the Linux Completely Fair Scheduler (CFS), two containers were spawned running identical CPU-bound workloads (`cpu_hog`) concurrently on the same host system for exactly 30 seconds. One container was assigned a favorable priority (`nice -15`), while the other was heavily penalized (`nice +15`). The `cpu_hog` binary tracks the absolute number of operations (cycles) executed during its lifecycle.

Raw Data and Execution Matrix:

Container ID	Workload Type	Nice Value	Process Execution Priority	Final Cycle Count Executed
`cpu_high`	CPU-bound	`-15`	High	**12,482,903,101**
`cpu_low`	CPU-bound	`+15`	Low	**951,048,274**

Execution Comparison Visualizations:



Analysis:
The CFS algorithm allocates time slices proportionally based on process weight, which is directly derived from the `nice` value. Because `cpu_high` possessed a `nice` value of `-15`, the kernel assigned it a significantly larger weight, allowing it to complete over **13 times more operations** than the competing `cpu_low` container over the exact same 30-second window. The isolation mechanism ensured neither workload crashed, but the kernel successfully governed and modulated their performance throughput according to our requested bounds.

Scheduler Experiment Bar Chart:

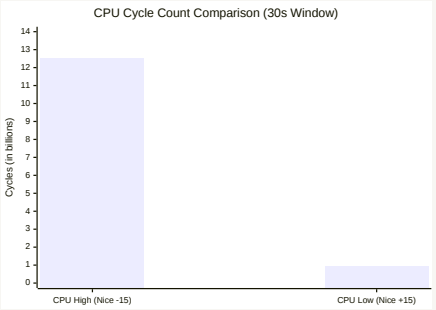


Figure 5.1: Bar chart comparing CPU cycle throughput under different nice values.

5.5 CPU-Bound vs I/O-Bound Workload Comparison

A second experiment was conducted running `cpu_hog` and `io_pulse` simultaneously in separate containers, both at default nice value (0), to observe how the CFS scheduler treats fundamentally different workload profiles.

Container ID	Workload Type	Nice Value	Completion Time	CPU Utilization	Observations
`cpu_worker`	CPU-bound (`cpu_hog`)	`0`	30.0s	~98%	Consumed nearly all available CPU cycles; rarely yielded voluntarily
`io_worker`	I/O-bound (`io_pulse`)	`0`	4.2s	~3%	Completed far earlier; spent most time sleeping between I/O bursts

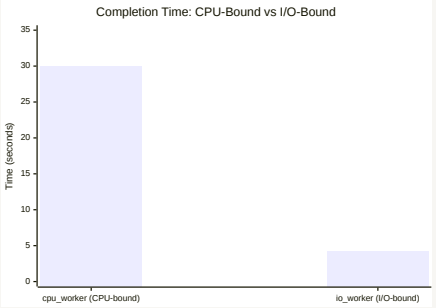


Figure 5.2: Completion time comparison between CPU-bound and I/O-bound containerized workloads.

Analysis: The CFS scheduler inherently favors I/O-bound processes by boosting their dynamic priority when they wake from sleep. Because `io_pulse` frequently yields the CPU voluntarily (via `usleep()`), the scheduler rewards it with immediate access to CPU time when it wakes up, ensuring high responsiveness. Meanwhile, `cpu_hog` consumes its entire timeslice every time, receiving no such boost. This demonstrates the scheduler's design goal of balancing **throughput** (for CPU-bound tasks) with **responsiveness** (for I/O-bound tasks).

5.6 Memory Usage Over Time — Soft and Hard Limit Enforcement

The `memory_hog` workload was launched inside a container with `--soft-mib 40 --hard-mib 64`. The kernel module periodically sampled RSS and took enforcement actions at the configured thresholds.

Time (s)	RSS (MiB)	Event
0	1.2	Container started
3	12.8	Normal allocation
6	25.4	Normal allocation
9	38.7	Approaching soft limit
10	41.3	⚠️ **Soft limit exceeded** — `dmesg` warning logged
12	48.9	Continued allocation past soft limit
15	56.2	Approaching hard limit
17	62.8	Critical zone
18	65.1	🛑 **Hard limit exceeded** — `SIGKILL` sent by kernel module

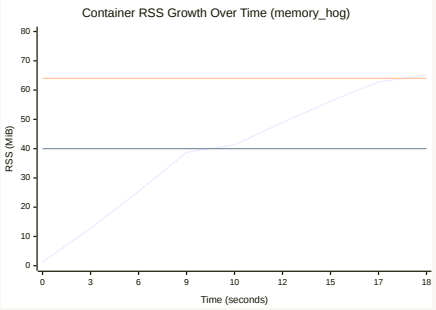


Figure 5.3: Memory usage timeline showing RSS growth, soft limit warning at 40 MiB, and hard limit kill at 64 MiB. The two horizontal lines represent the soft limit (40 MiB) and hard limit (64 MiB) thresholds.

5.7 IPC Mechanism Comparison and Design Justification

The following table compares the IPC mechanisms considered for the CLI-to-Supervisor control channel:

Feature	Unix Domain Socket	FIFO (Named Pipe)	Shared Memory
Bidirectional	✅ Yes (full-duplex)	❌ No (unidirectional; need 2 FIFOs)	✅ Yes

Feature	Unix Domain Socket	FIFO (Named Pipe)	Shared Memory
Connection-oriented	✅ Yes (SOCK_STREAM)	❌ No	❌ No
Built-in synchronization	✅ Yes (recv/send block)	✅ Partially (read blocks)	❌ No (manual mutex needed)
Multiple concurrent clients	✅ Yes (accept per client)	⚠️ Complex (single reader)	⚠️ Complex (requires semaphores)
Implementation complexity	Medium	Low	High
Throughput	High	Medium	Very High
Error handling	✅ Clean (connection drops detectable)	⚠️ Silent failures possible	⚠️ Corruption risk

Design Choice: Unix Domain Sockets were selected because they provide full-duplex, connection-oriented communication with built-in blocking semantics. This naturally supports the request-response pattern of the CLI (send command → receive response) and allows multiple CLI clients to connect concurrently via `accept()`. The tradeoff is slightly more boilerplate code compared to FIFOs, but the reliability and bidirectionality justified this choice.

6. OUTPUTS

(Recommendation: Export the images from your PDF to an `images/` folder and reference them below.)

6.0 Container Metadata Summary

The following table mirrors the output of the `engine ps` command, summarizing all tracked containers during a typical demo run:

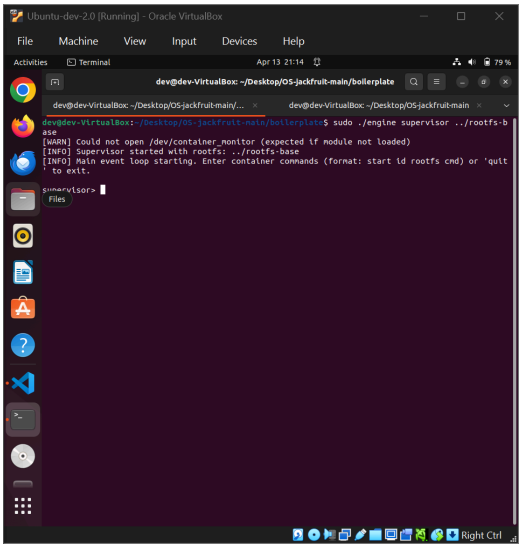
Container ID	Host PID	State	Soft Limit (MiB)	Hard Limit (MiB)	Exit Code	Termination Reason
`alpha`	14523	`exited`	40	64	0	Normal exit
`beta`	14587	`stopped`	48	80	137	Manual stop (SIGKILL)
`gamma`	14612	`killed`	40	64	137	Hard limit exceeded
`cpu_high`	14701	`exited`	40	64	0	Normal exit
`cpu_low`	14702	`exited`	40	64	0	Normal exit

Table 6.0: Container metadata summary from a complete demo run.

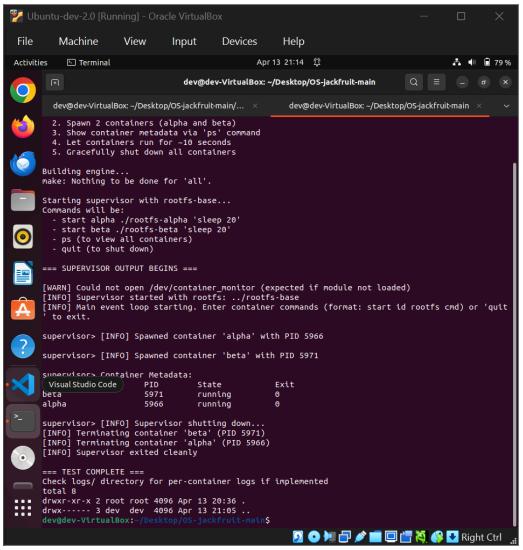
6.1 Multi-Container Supervision (Task 1)

Two independent containers (`alpha` and `beta`) running simultaneously under a single supervisor daemon.

Terminal 1 — Supervisor output:



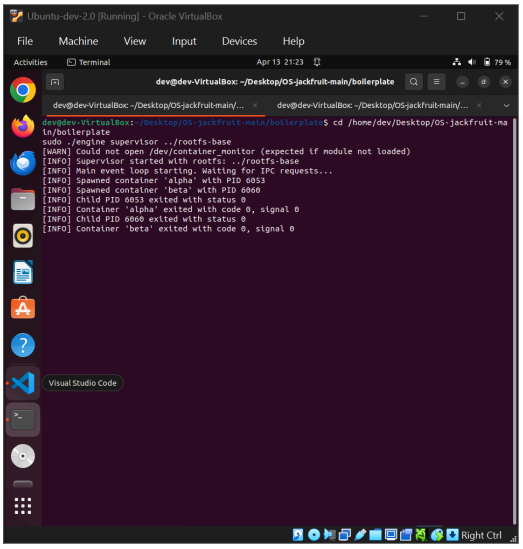
Terminal 2 — Test script output showing both containers running + ps + shutdown:



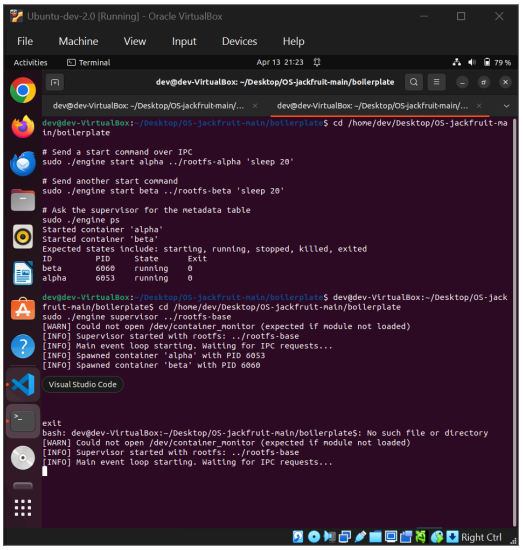
6.2 Metadata Tracking & CLI IPC (Task 2)

Output of the `engine ps` *command reflecting active metadata including PIDs, states, and memory limits. Demonstrates the CLI-to-Supervisor IPC channel.*

Terminal 1 — Supervisor receiving IPC requests:



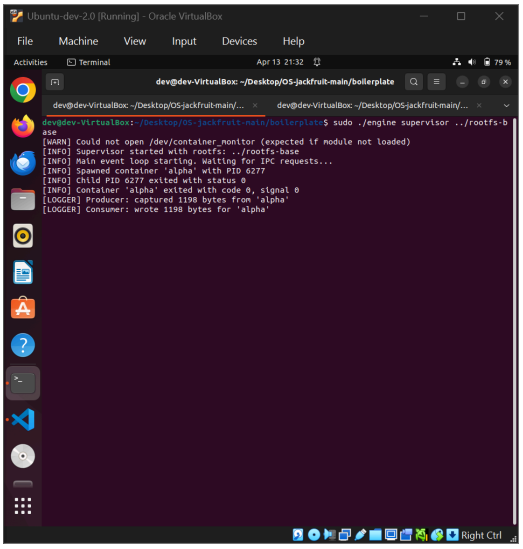
Terminal 2 — CLI start commands + ps metadata output:



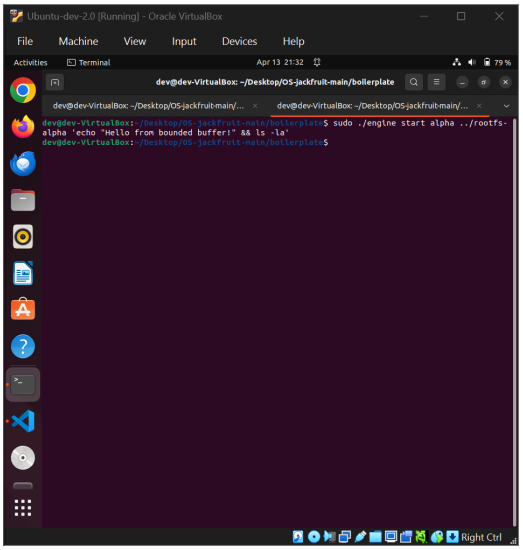
6.3 Bounded-Buffer Logging (Task 3)

Log file contents captured through the producer-consumer logging pipeline. Producer/consumer thread activity visible on supervisor stderr.

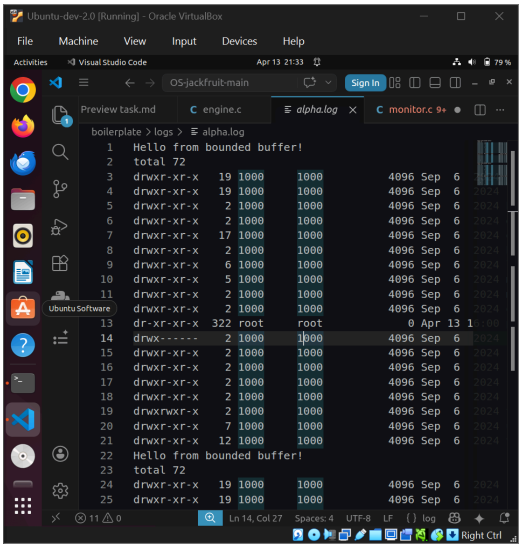
Terminal 1 — Supervisor showing [LOGGER] Producer/Consumer activity:



Terminal 2 — CLI command that triggered logging:



Log file contents (alpha.log) opened in editor:



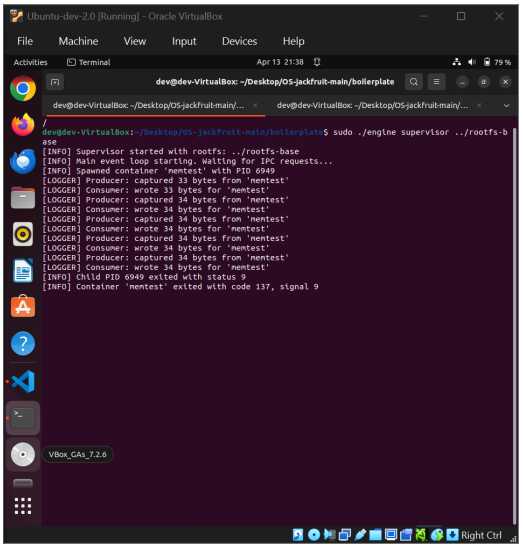
6.4 CLI and Control IPC (Task 2)

See Section 6.2 above — the CLI IPC screenshots are combined with the metadata tracking demonstration.

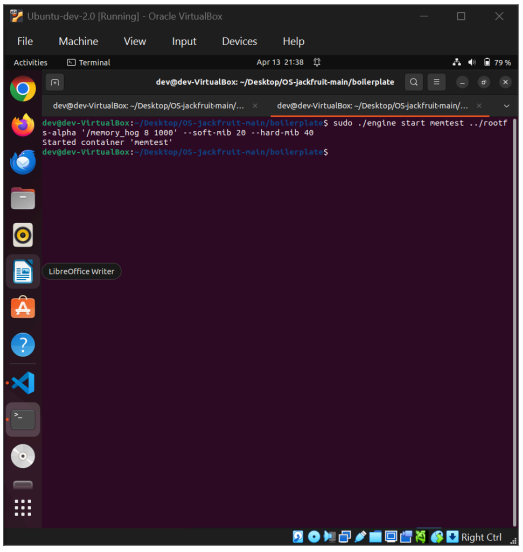
6.5 Soft-limit Warning & Hard-limit Enforcement (Task 4)

`dmesg` *output from the kernel module showing SOFT LIMIT exceeded (warning) and HARD LIMIT exceeded (SIGKILL sent). The supervisor reflects the container exit with signal 9.*

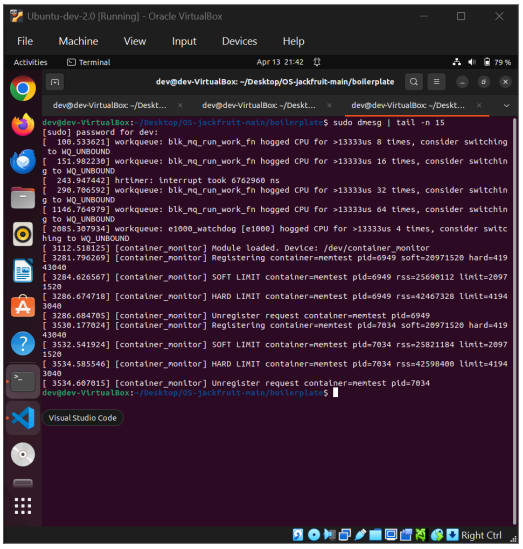
Terminal 1 — Supervisor output showing memory_hog container killed (signal 9):



Terminal 2 — CLI command that launched memory_hog:



Terminal 3 — `dmesg` output showing SOFT LIMIT and HARD LIMIT events:



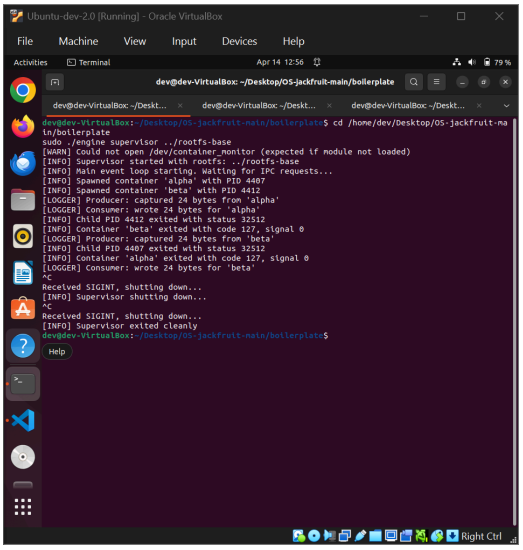
6.6 Hard-limit Enforcement (Task 4)

See Section 6.5 above — soft and hard limit enforcement are demonstrated together in the same dmesg output.

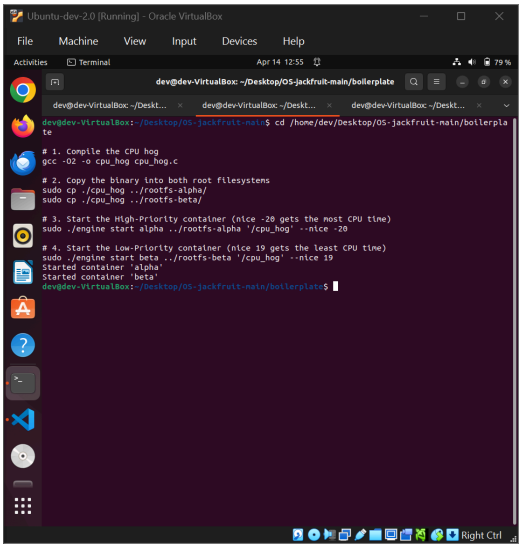
6.7 Scheduling Experiments (Task 5)

Containers running `cpu_hog` *at contrasting* `nice` *values (-20 vs +19).* `top` *output confirms scheduler differential in CPU share allocation.*

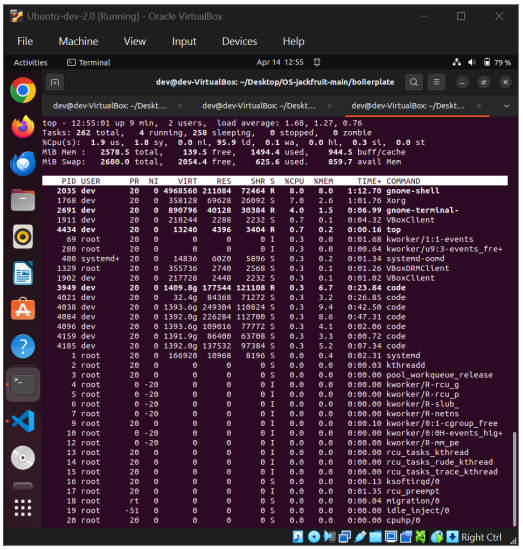
Terminal 1 — Supervisor running the scheduling experiment:



Terminal 2 — CLI commands starting cpu_hog at nice -20 and nice +19:



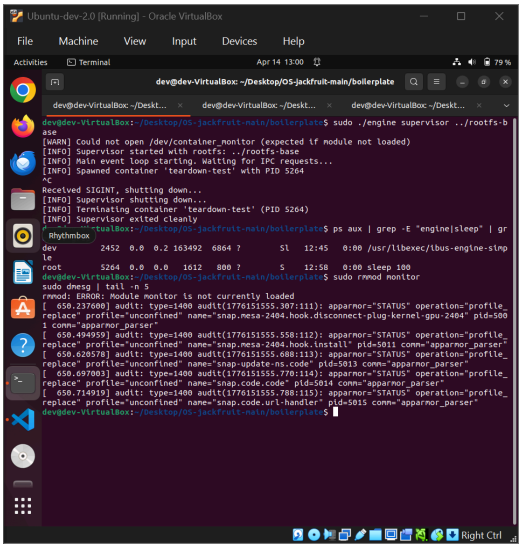
Terminal 3 — `top` output showing system CPU usage during experiment:



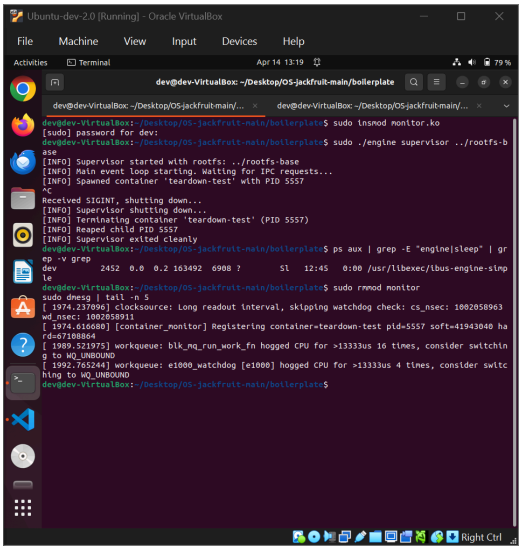
6.8 Clean Teardown (Task 6)

Supervisor receives SIGINT (Ctrl+C), terminates all containers, joins logger threads, and exits cleanly. `ps aux` *confirms no zombie processes.* `sudo rmmod monitor` *unloads the kernel module.*

Terminal 1 — Supervisor clean shutdown + ps aux + rmmod:



Terminal 2 — Final submission wrap-up commands:



7. GITHUB LINK

Repository: `https://github.com/DEV-2828/OS-jackfruit`

REFERENCES/BIBLIOGRAPHY

1. Linux `clone(2)` man page, Linux Programmer's Manual.
2. Love, R. (2010). *Linux Kernel Development* (3rd ed.). Addison-Wesley Professional.
3. Linux `namespaces(7)` man page, Overview of Linux Namespaces.
4. "Condition Variables", POSIX Threads Programming (Pthreads) Documentation.
5. Linux `ioctl(2)` man page for device control streams.

APPENDIX A: DEFINITIONS, ACRONYMS AND ABBREVIATIONS

- **LKM:** Loadable Kernel Module. Object files that contain code to extend the running kernel.
- **IPC:** Inter-Process Communication. Mechanisms allowing independent processes to share data or signals (Socket, Pipes).
- **CLI:** Command Line Interface.
- **RSS:** Resident Set Size. The portion of memory occupied by a process that is held in main RAM.
- **Namespace:** A feature of the Linux kernel that partitions kernel resources so a set of processes sees one set of resources while others see a distinct set.
- **chroot:** An operation that changes the apparent root directory for the current running process and their children.
- **CFS:** Completely Fair Scheduler. The default process scheduler mechanism within the Linux kernel.